

The Post-API Age

Week 3 · Foundations module · Textbook Chapter 3

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This week at a glance

The web used to invite us in. That began to change in recent years.

Why did this bargain change? *How* can we keep doing web data science anyway?

Monday | Concepts

- The golden age
- Enclosure, exemption & erosion
- Counter-values, lineages & misconceptions

Wednesday | Notebook lab

- Measure data access and forensic analysis of dead endpoints

Friday | Textbook revisions

- Reviewing issues and proposing pull requests

Monday lecture

Read Chapter 03 of the Web Data Science book

Wednesday lab

Complete **Recommended Exercises**: fill in the empty cells, run everything, export to HTML, upload to Canvas by **Sunday 11:59pm**

Friday notebook

You do not need to wait until Friday to begin to submit issues from textbook or notebook, upload link to issue by **Sunday 11:59pm**

1. Monday • Concepts

The golden age of web data access

Between roughly 2008 to 2018 the web was a mostly open that was easy(ish) to retrieve and analyze

Websites rendered a mix of static and dynamic content

Social platforms shared data: Twitter in 2006; Facebook in 2010; Reddit's API was free; Google exposed trends

Platforms traded data access for lock-in to their ecosystems, credibility, recruitment

This built my field of computational social science and a generation of data journalism and civic tech

This is the “API age” → we’re now in the “post-API age”



They just gave out data for free!

Three pressures

(1) Enclosure, (2) Exemption, (3) Erosion

Why platforms say no

Platforms invoke multiple reasons for restricting researcher access

- **Privacy & safety.** User data can expose identities, enable harassment, or be repurposed in ways users never expected
- **Infrastructure.** Scraping and bulk API access impose real engineering and moderation costs
- **Business.** Data are an asset: competitors, AI companies, and researchers all want something the platform paid to collect
- **Legal risk.** Copyright, privacy law, contracts, and regulatory obligations make broad access risky
- **Control.** Platforms cannot easily distinguish public interest research from spam, surveillance, or commercial extraction

The platform's questions

Who counts as a researcher?

Access to what data?

Under what rules?

How does it help us?

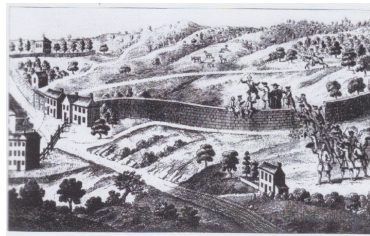
Pressure 1 — Enclosure

Enclosure privatizes a shared resource — traces back to the fencing in of English common land. The data still exist, just on the other side of a paywall.

- **Twitter/X**: a free academic archive (2006–2021) → discontinued; enterprise access starts at \$42,000/month
- **Reddit**: free since 2008–2023 → \$0.24 per 1,000 calls; 7,000+ subreddit protests and Pushshift cut off

Two drivers:

1. **Independent scrutiny** — researchers and journalists answering uncomfortable questions about polarization, harassment, privacy, financials, *etc.*
2. **AI training data** — uncontaminated user-generated content becomes enormously valuable for training AI models



This land is my land now, it isn't your land anymore

Exemption: platforms operate without transparency or accountability while surveilling and experimenting on you.

From last week: when “research” appears in a TOS or Privacy Policy, it is generally permission for *them* to do it to *you*

Researchers who want to hold platforms accountable get:

- cease-and-desist letters citing the Computer Fraud and Abuse Act
- NDA'd research collaborations
- data transparency delayed for years
- “privacywashing” and “ethicswashing”



[Back to Newsroom](#)

META

Research Cannot Be the Justification for Compromising People's Privacy

August 3, 2021

By Mike Clark, Product Management Director

Meta is suddenly very worried about privacy when researchers measure Meta behaving badly

Erosion is the degradation in technology, practices, and other infrastructure that enables web data science.

- **Links rot.** About half the links in U.S. Supreme Court opinions are dead
- **Platforms churn.** GeoCities vanished; Parler disappears; Twitter became X
- **Sites redesign** and silently break every scraper

For research this is a **reproducibility crisis**: data cannot be re-collected and analyses cannot be re-run → is it still science if it just worked one time for one team?

...sylvania (with a population of about 12.8 million adds 1.4 million more; and New York City 8.4 million) adds another 1.2 million. See ...; Associated Press, Pa. Database, NBC News ...), online at <http://goo.gl/3Yq3Nd> (as last 17, 2016); N. Y. Times, Oct. 8, 2015, p. A24.¹ ...ng warrants do not appear as bolts from the re the run-of-the-mill results of police stops— ... look for when there was a routine check of a

SCOTUS decisions using broken links

Openness: conduct your own work so that it resists enclosure

Document *how* you collected: endpoints, parameters, dates, and the platform's constraints at collection time

This documentation lets a future researcher understand your dataset after the source changes or vanishes

Practice redundancy: deposit data in durable repositories (Zenodo, Dataverse, etc.), lean on the Internet Archive and Common Crawl, prefer openly licensed sources

Wikipedia as the durable counter-factual — open API, open license, full public dumps

Openness in one habit

Record the date. Preserve your data.
Repeat to document changes.

Oversight: consequential systems should be inspected by parties who do *not* own them.

Independent audits of platform algorithms and content moderation; documentation standards that make datasets and methods legible to review

Regulation converts access from a favor into an obligation: EU DSA Article 40 compels vetted-researcher access to very large platforms

This course is oversight

Someone who can retrieve, parse, and analyze web data independently is someone who can hold platforms accountable instead of asking politely

Ownership: prefer arrangements where communities govern the infrastructure their data lives in rather than trusting someone else will keep it running forever

Federated systems on open protocols distribute control: no single acquisition or pricing decision can enclose the whole network

- HTTP, torrents, RSS, Mastodon, Bluesky, *etc.*

Community archives, data cooperatives, and volunteer efforts hold collective traces collectively

The trade

Federation exchanges convenience for control: smaller populations, uneven moderation, more work per query

Five common beliefs. For each, commit: **true or false?**

1. “The post-API age means APIs are gone.”
2. “If the API closes, I’ll just scrape.”
3. “If data is publicly, collecting it is permitted.”
4. “This only matters for social media research.”
5. “Nothing can be done about it.”

Rules

Write your five answers down first; then we take a hand count, myth by myth. Defend your minority votes — some of these are *almost* true.

1. **APIs are everywhere.** More exist than ever; what is disappearing is *free access for outside observers*, replaced by paid tiers, gated programs, and licensing deals
2. **Scraping is one tool in a portfolio.** Legal risks are changing but non-zero, adversarial site design, poor scalability
3. **Visibility, permission, and ethics are three different questions** A public post's author may never have expected a dataset; a viewable page's ToS may forbid automated collection
4. **Enclosure follows AI-training value** wherever it appears: news archives, code repositories, forums, image libraries. Government data stays open by legal mandate
5. **The environment is contested.** Regulation is creating access obligations, archives are institutionalizing preservation, and federated alternatives are growing

Other professions and disciplines

Public interest data science asks how technical skills can increase the public's capacity to know, scrutinize, and act on consequential institutions and systems.

Retrieving data is never only a technical problem:

- **Access** — who can observe powerful institutions
- **Methods** — what can be made visible and auditable
- **Infrastructure** — whether evidence remains available
- **Practice** — who benefits from our technical work

Other professions have met the problem of powerful actors controlling information the public needs.

Five older traditions

Other flavors of public interest

Journalism — scrutiny

Law — compelled disclosure

Accounting — independent audit

Urban planning — participation

Engineering — public safety

The point

Not one definition of “the public good,” but traditions for making power observable, reviewable, and contestable.

Journalism is not just inspiration for public interest data science; it is an older professional technology for working with contested evidence.

FOIA and state records laws turn watchdog work from a polite request into a procedural right: deadlines, exemptions, appeals, and records that agencies did not voluntarily publish.

Verification and provenance are default practices: a claim must trace back to a source, a document, a row, or an interview.

Right of reply treats the target of an investigation as part of the verification process, not as a veto-holder.

Collaborative investigations scale scrutiny beyond any one newsroom.

Data science echo

Journalism's lesson is procedural:

show your work, preserve provenance,
invite challenge, publish under pressure

Law contributes procedures for extracting evidence from institutions that have incentives not to disclose it.

Public-records law creates a statutory geometry: request, deadline, exemption, appeal, litigation.

Discovery compels records, testimony, and interrogatory answers that no public API or FOIA request would expose.

Cause lawyering treats a case as a way to change a rule, not merely resolve one dispute.

Sandvig v. Barr matters because it reduced the CFAA chill around audit studies that use bots, paired accounts, or ToS-violating methods.

Data science echo

Legal infrastructure before technical accountability can begin:

requests, complaints, briefs, subpoenas, protective orders.

Accounting contributes the machinery for making claims credible to other institutions.

Audits create an independent record of what was checked, how, and against which standards.

Materiality distinguishes errors that matter from noise: not every discrepancy changes the public claim.

Workpapers make the audit inspectable: evidence, judgment calls, sampling choices, and reviewer sign-off.

Standardized reporting makes one organization's claims comparable to another's.

Data science echo

Datasheets, model cards, replication packages, pinned environments, and audit logs are early versions of inspectability.

Affected publics: technical decisions are also decisions about land, exposure, mobility, housing, and political voice.

Environmental impact statements force agencies to document consequences before irreversible action.

Public comment creates a record, but not necessarily power: consultation can become participation theater.

Stakeholder maps make visible who is in the room, who is affected, who has formal authority, and who has remedy.

Data science echo

Disclose before deployment, map affected parties, document expected harms, and create a record for challenge.

Technical practitioners have public duties that can override duties to clients or employers.

Licensure gives some practitioners standing to refuse unsafe work.

Codes and standards convert past failures into shared constraints on future design.

Ethics canons name the public as a client even when the public did not sign the contract.

Failure analysis treats accidents as institutional evidence with required investigation and disclosure

Data science echo

Data science has no equivalent license to lose, no universally enforced canon, and no mature failure-reporting regime.

Researchers have challenged the idea that platform terms alone should define the boundary of legitimate inquiry.

- **Sandvig v. Sessions/Barr.** Civil-rights researchers challenged CFAA risk around audit studies using bots, scraping, and tester accounts
- **Van Buren aftermath.** Terms-of-service violations alone are a weaker basis for treating research as “hacking”
- **Strategic target.** The point is reducing the legal chill around public interest auditing

The legal move

Can researchers test platforms the way civil-rights testers test landlords, lenders, or employers?

Pushback 2 — Build around the missing API

Build research capacity outside of platforms with technical infrastructure you control

- **Platform research tools.** Meta Content Library/API, TikTok Research API, controlled-access clean rooms
- **Independent archives.** Internet Archive, Common Crawl, End-of-Term archives, local WARC collections
- **User-side collection.** Data donations, browser instrumentation, panels, screen capture, audit accounts
- **Shared infrastructure.** Replication packages, codebooks, documented missingness, synthetic examples

Major trade-offs with comprehensiveness, reliability, resolution

New access model

Move from a single point of failure to archives + enclaves + donations + audits

Pushback 3 — Turn access into a policy problem

Researchers also push by making platform access a matter of public obligation, not platform generosity.

- **EU: DSA Article 40.** Vetted researchers can seek public or non-public VLOP/VLOSE data
- **Federal proposals.** Researcher requests require ad libraries, moderation stats, and recommender data
- **State transparency laws.** California AB 587 seek disclosure, but 1A challenges narrow what states can compel
- **Public officials on private platforms.** FOIA, state public-records laws, and 1A doctrine can turn official accounts into transparency targets

Public power in private infrastructure still deserves public oversight.



Activity — Platform vs. researcher hidden profile negotiation

Researchers

If your birthday is an even day, you are a naughty researcher trying to get platform data for evil

If your birthday is an odd day, you are a nice researcher trying to get platform data for good

Records — posts, users, comments, ads, algorithms?

Fields — text, timestamps, engagement, deletions?

History — how far back? how often?

Platform operators

If your birthday is an even day, you are a naughty platform trying to protect platform data for evil

If your birthday is an odd day, you are a nice platform trying to protect platform data for good

Privacy — could users be identified or harmed?

Revenue — are you giving away a valuable asset?

Infrastructure — what does bulk access cost?

Competition — could others copy products or models?

Legal risk — copyright, contracts, regulators, lawsuits?

Enclosure vs. openness; Exemption vs. oversight; Erosion vs. ownership and appeals to public interest

Debrief — What did access cost?

Summarize your deal.

1. What did the researchers originally ask for?
2. What did the platform actually agree to provide?
3. What restriction was hardest to negotiate?
4. What did you trade away to get access?
5. Reproducible by other parties?

Researchers	Nice	Resistance legitimate research barred	Cooperation responsible access
	Naughty	Extraction abuse, spam, surveillance	Refusal protecting users
		Naughty	Nice
		Platforms	

Wednesday: measure it, then take it home

Build an access matrix: probe eight live endpoints with no credentials and tabulate who answers, who refuses, and how

Run dead-endpoint forensics: autopsies of Pushshift, CrowdTangle, and Twitter v1.1 — three retirements, three different failure signatures

Take-home: the 7-step audit

1. Named `User-Agent`
2. Probe one endpoint by hand
3. All eight → dated DataFrame
4. Add two endpoints *you* care about
5. Diagnose one refusal, attribute it
6. Autopsy one retired endpoint
7. Write up what changed

2. Wednesday · Notebook Lab

Claims you can check — the access matrix

What happens if you show up to a website with no credentials and ask for data?

```
def probe(name, url):  
    """Ask one endpoint for data without credentials; describe the answer."""  
    try:  
        response = requests.get(url, headers=HEADERS, timeout=20)  
    except requests.RequestException as error: # no answer is a result too  
        return {"api": name, "status": None, "verdict": "unreachable"}  
    verdict = ("open" if response.status_code == 200  
              else "gated" if response.status_code in (401, 403) else "other")  
    return {"api": name, "status": response.status_code, "verdict": verdict,  
           "content_type": response.headers.get("content-type", "")}
```

Measured 28 Aug 2026: Wikipedia, Hacker News, arXiv, Mastodon, Bluesky **open**; Reddit and X **gated**; Meta Graph refuses without a token. Your run *will* differ — that is the finding, so record the date.

Read the rows, not the totals

Wikipedia, Hacker News, arXiv answer anonymous requests with data, and have for a decade — a non-profit, an incubator, a university consortium

Reddit: status 403, content-type `text/html` — not an API refusing you, a *block page* served where JSON lived from 2008 to 2023

X: 401 `application/problem+json` — a machine-readable refusal



Dead-endpoint forensics — three ways to die

How a retired endpoint refuses tells you whether anything can be recovered:

- **Pushshift** answers 403 in JSON: `{"detail": "Not authenticated"}`. **Gated, not gone** — it still runs, for Reddit moderators. A gated service can be negotiated with.
- **CrowdTangle** does not resolve at all. No HTTP conversation happens. **Enclosure completed**; nothing left to ask.
- **Twitter v1.1** parses your request and rejects it (error 215). Alive, serving paying customers. **Repriced, not removed**.

The archive keeps the receipts

Wayback's last captures: `api.pushshift.io` — 25 May 2023, six weeks before Reddit's pricing took effect. `crowdtangle.com` — 14 Aug 2024, the month Meta retired it. The shutdowns are written in the archival record.

Archived docs are how you reconstruct what a five-year-old paper's dataset contained. More in Ch. 7.

The take-home — a guided audit in seven steps

The notebook's **Recommended Exercises** walk the audit end to end. Fill in each empty cell:

1. Define **HEADERS** with your named User-Agent
2. Probe **one** endpoint by hand; look at the answer
3. Build the eight-endpoint matrix as a DataFrame
4. Add **two** endpoints you might actually study
5. Diagnose one refusal: **server**, **via**, body
6. Autopsy one retired endpoint: gone, gated, or repriced?
7. Interpret: what differed from the book, and why?

Submitting

Run all cells, then export: *File* → *Save and Export Notebook As* → *HTML*. Upload the `.html` file to Canvas by **Sunday 11:59pm**. Canvas accepts only `.html` — code and output together, as executed.

Start in class, finish at home. Work in pairs; submit your own run.

If you finish early — the additional exercises

The chapter's **Additional Exercises** are optional, open-ended extensions — worth your time, not required:

1. **Extend the autopsy** — two more retired endpoints: gone, gated, or repriced, and who is left to ask?
2. **Timeline** — five API-access events, three platforms, 2008 to today; tag each with its pressure.
3. **Framework application** — 500 words on a platform change *not* in the chapter.
4. **Contingency plan** — a project on one source, plus three fallbacks if it vanishes mid-study.
5. **Public interest audit** — how would current restrictions hit a real API-enabled project?
6. **5617**: read Davidson et al. (2023) and Perriam et al. (2020); write a 2–3 page research-access brief for one platform.

Fallbacks worth knowing

Official research programs (real but rationed) · data donation · FOIA / records requests (Ch. 11) · crowdsourcing.

Show-and-Tell

Bring one: an access change that surprised you, a dataset you fear losing, or a question for a research design.

The audit teaches you to date-stamp a measurement; openness generalizes that habit. Record **what** you collected and also **how**: endpoints, parameters, dates, and the platform's constraints (Gebru et al. 2021). A future reader (or you, in a year) can then reconstruct exactly what was allowed:

```
collection = {
    "source": "reddit.com (PRAW, r/ClimateChange)",
    "endpoint": "GET /r/{sub}/top params: t=year, limit=1000",
    "collected": "2026-02-14",
    "access_tier": "researcher program (application-gated)",
    "constraints": "no redistribution of raw comments; 100 req/min",
    "known_gaps": "pre-2023 history unavailable -- Pushshift offline",
}
import json, pathlib
pathlib.Path("datasheet.json").write_text(json.dumps(collection, indent=2))
# The datasheet travels next to the data; the source may change, the record does not.
```


“If the API closes, I’ll just scrape.”

Visibility, permission, and ethics are three different questions.

Scraping is one tool in a portfolio —
not an escape hatch that makes enclosure irrelevant.

3. Friday • Textbook Revisions

Improving Chapter 3 together

Last week you filed an **issue** — a report that something is wrong. Today you take the next step and propose the **fix** itself: your **first pull request**, against `ch-03-post-api.qmd`, plus reviews of two classmates’.

The workflow, in GitHub Desktop or `gh` (no raw `git` needed):

1. **Clone** the book repo, then create a **branch**
2. Edit the chapter; **commit** with a clear message
3. **Push**, then open a PR describing *what* and *why*
4. Review two peers’ PRs: kind, specific, actionable

This is a **fast-moving** chapter — prices, lawsuits, and regulations change monthly — so it always needs hands.



A revision menu for Chapter 3

High-value targets — pick one and scope it to a single PR:

- **Add the missing figures.** The chapter still has no figures. Contribute a three-pressures diagram, an API-access timeline (2008–2026), or a link-rot chart — the deck’s placeholders show the need.
- **Re-run the audit.** The access matrix and forensics outputs are dated 28 Aug 2026 and *will* drift. Re-run the chapter’s code, and where a row changed, update the printed output and the prose that reads it. This is the chapter’s built-in revision engine.
- **Update the moving numbers.** Verify and re-cite the Twitter/X pricing, the Reddit–Google deal, and DSA Article 40’s implementation status “as of mid-2026.” Flag what has gone stale.
- **Resolve the editorial TODOs.** The source carries author-voice and “add citation when published” notes (the openness/oversight/ownership framing). Track down a source or draft the text.
- **Extend a Misconception or Further Reading.** Add a post-2023 source on AI-training licensing, tied to “enclosure logic is spreading.”
- **Verify the cross-references.** Ch. 3 leans on many @sec- links (ethics, wikipedia, archives, automation, social, government). Confirm each resolves and is described accurately.

A good revision, and a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, source)
- Separates “must fix” from “nice to have”
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Data formats** — XML & JSON, the shapes web data actually arrives in.

Key takeaways

1. The open-API era ($\sim$ 2008–2018) was a **strategic bargain**, not a natural state — and computational social science was built on the trade.
2. **Enclosure** converts collective data into private assets: shutdowns, repricing, and AI-training licensing deals.
3. **Exemption** is the asymmetry of scrutiny — platforms study users freely while blocking those who study platforms.
4. **Erosion** is infrastructural decay, so reproducibility is a problem you must solve *at collection time*.
5. **Access is measurable, not just arguable** — an unauthenticated probe places a platform on the open-to-closed gradient today, and *how* a dead endpoint refuses tells you whether anyone is left to negotiate with.
6. The response is **redundancy**: openness, oversight, ownership — and mastery of many access techniques so no single closure ends your research.